

# Bayesian Modeling

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“We are all Bayesians at heart — in that we all have prior knowledge — so why use a frequentist approach to statistics?”

**John J. McArdle**

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# Bayesian Modeling

## CFA & SEM

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## Concepts

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## Vocabulary

- ❑ **Factor analysis:** refers to a set of latent variable models and procedures for fitting them to data
- ❑ **Factor/latent variable:** continuous construct that cannot be observed directly (like anxiety)
- ❑ **Manifest/observed variable:** observed directly (measured variable, indicator)

## Generalization of common models

- ❑ **Generalized Linear Models:** measurement error not modeled; all variables treated as manifest even when imperfectly measured constructs are of interest
- ❑ **Constructs** are imperfectly measured because: they are somewhat abstract, there are obstacles to accurately measure

## Factor analysis

- ❑ **Factor analysis** is used to study the structure underlying data in pursuit of a simpler representation (dimension reduction)
- ❑ **Objective:** identify the number and nature of the factors that produce the observed relations between manifest variables
- ❑ Assume: fewer factors than manifest variables, a limited number of fundamental constructs explain interrelationships between all manifest variables

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## Common uses

- ❑ Though there are many potential uses of factor analysis, it is often used to create, refine, or evaluate scales, tests, questionnaires, interviews, etc.
- ❑ Identify unidimensional constructs measures by a multidimensional instrument to create subscales
- ❑ Evaluate validity of a scale according to which manifest variables load on the same factor(s)

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## Partitioning variability

- ❑ Observed variability in each manifest variable is presumed due to common variance and unique variance
- ❑ **Observed = common + unique**
- ❑ **Common variance:** shared with other manifest variables, common to more than one manifest variable
- ❑ **Unique variance:** is not shared with other manifest variables
- ❑ **Reliability:** proportion of total observed variability that is not error

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## EFA – Exploratory factor analysis

- ❑ Exploratory because the factor structure in the data is not clear in advance of an analysis (sometimes not after the analysis either).
- ❑ Often called “unrestricted” because all manifest variables can be influenced by all factors
- ❑ Replication with separate data is essential

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## PCA – Principal component analysis

- ❑ Data reduction procedure that does NOT involve latent variables. It is not a probabilistic model, just a linear combination/transformation of observed variables
- ❑ When the goal is to understand underlying dimensions implied by correlations between manifest variables, and interpret the dimensions as constructs; FA is applicable and PCA is NOT

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## CFA – Confirmatory factor analysis

- ❑ Confirmatory because a particular factor structure is hypothesized and tested. Alternative models with different factor structures may be compared
- ❑ CFA is a subset of SEM (structural equation models). **A SEM consists of a measurement model plus relations between latent variables**
- ❑ CFA uses the same model as EFA (with different constraints)

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## IRT – Item response theory

- ❑ A set of latent variable models and procedures for fitting them to data. Used primarily for scale development, refinement, and scoring
- ❑ Almost all **IRT models** are for binary, nominal or ordinal manifest variables. Traditional **factor analysis** is for interval and ratio manifest variables
- ❑ The factor analytic models that are equivalent to IRT are specially designed for categorical manifest variables

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## Latent variable models

- ❑ A model is fitted to data and **all models are wrong to some degree**, the data may not be explained perfectly
- ❑ Interpretations must involve a subjective component and solutions will not make sense
- ❑ Model fit should be evaluated
- ❑ Latent variable models with continuous latent variables can take different forms – EFA, CFA, IRT models – all examples of a multivariate (generalized) linear model

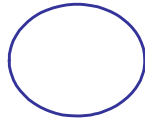
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## Conceptual conventions



**Circles/Ovals: Latent Constructs**  
[Circles are also used for residual variables]



**Boxes/Rectangles: Manifest (measured) variables**



**Single-headed lines: Directional (causal) relationships represented as regression (slopes) estimates**



**Double-headed lines: Non-directional (co-occurring) relationships represented as covariance or variance estimates**

## Our first model

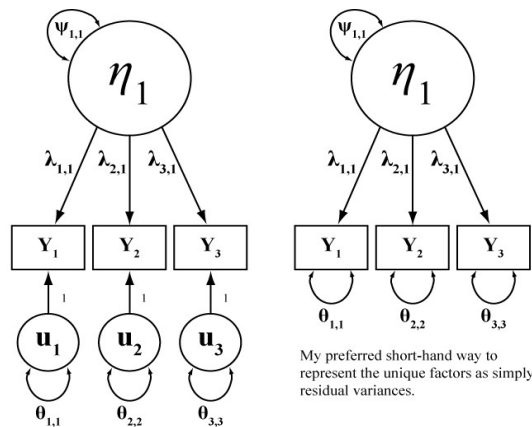


## First model - Data

|       |          |       |     |      |         |
|-------|----------|-------|-----|------|---------|
| Great | Cheerful | Happy | Sad | Down | Unhappy |
| 1     | 2        | 3     | 4   | 5    | 6       |

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## First model - Notation

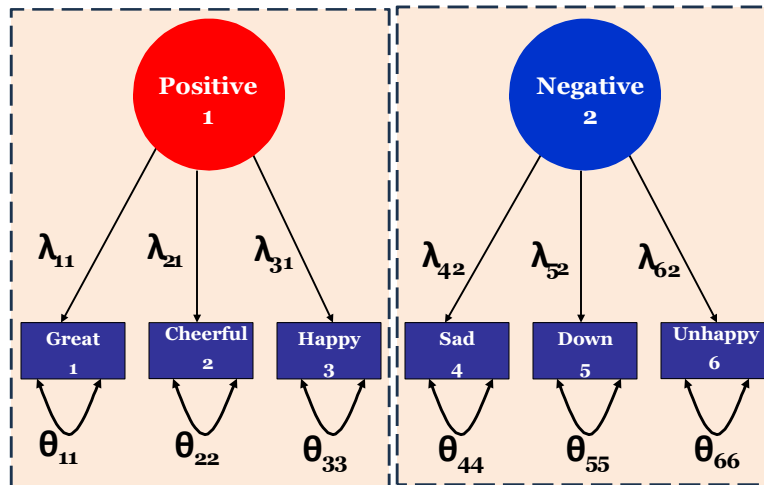


**Figure 3.2** More drawing conventions for SEM: A) the technically accurate way and B) a simpler short-hand way.

*Note.* Both drawings are identical models mathematically and, pretty much, conceptually. On the left, residual-variance information is treated as a unique factor with some estimated variance, while on the right, the residual-variance information is represented as ‘unexplained’ variance in the indicator. Unexplained variance is simply a unique factor.

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## First model – Measurement model



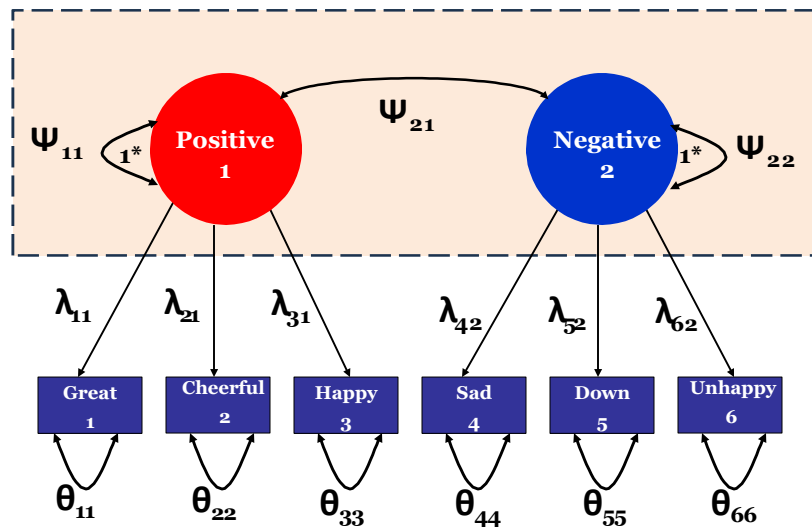
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## First model – Structural model



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## The factor model

- The equations of the model:

$$Y_{i1} = \mu_1 + \lambda_{11}Z_{i1} + u_{i1}$$

$$Y_{i2} = \mu_2 + \lambda_{21}Z_{i1} + u_{i2}$$

$$Y_{i3} = \mu_3 + \lambda_{31}Z_{i1} + u_{i3}$$

$$Y_{i4} = \mu_4 + \lambda_{42}Z_{i2} + u_{i4}$$

$$Y_{i5} = \mu_5 + \lambda_{52}Z_{i2} + u_{i5}$$

$$Y_{i6} = \mu_6 + \lambda_{62}Z_{i2} + u_{i6}$$

## Assumptions

- Everything is normally distributed (except with categorical data)
- Positive and Negative affects are multivariate normal; this is what we mean when we allow them to correlate
- ML estimation is correct asymptotically (large enough sample)
- Assuming at least centered data,  $\mu$  are null. Then, all information is reduced into the covariance matrix
- There is no prior knowledge in ML

## Implied covariances

| Great<br>1 | Cheerful<br>2 | Happy<br>3 | Sad<br>4 | Down<br>5 | Unhappy<br>6 |
|------------|---------------|------------|----------|-----------|--------------|
|------------|---------------|------------|----------|-----------|--------------|

P1:  $\lambda_{11}\Psi_{11}\lambda_{11} + \theta_{11}$

P2:  $\lambda_{11}\Psi_{11}\lambda_{21}$        $\lambda_{21}\Psi_{11}\lambda_{21} + \theta_{22}$

P3:  $\lambda_{11}\Psi_{11}\lambda_{31}$        $\lambda_{21}\Psi_{11}\lambda_{31}$        $\lambda_{31}\Psi_{11}\lambda_{31} + \theta_{33}$

N1:  $\lambda_{11}\Psi_{21}\lambda_{42}$        $\lambda_{21}\Psi_{21}\lambda_{42}$        $\lambda_{31}\Psi_{21}\lambda_{42}$        $\lambda_{42}\Psi_{22}\lambda_{42} + \theta_{44}$

N2:  $\lambda_{11}\Psi_{21}\lambda_{52}$        $\lambda_{21}\Psi_{21}\lambda_{52}$        $\lambda_{31}\Psi_{21}\lambda_{52}$        $\lambda_{42}\Psi_{22}\lambda_{52}$        $\lambda_{52}\Psi_{22}\lambda_{52} + \theta_{55}$

N3:  $\lambda_{11}\Psi_{21}\lambda_{62}$        $\lambda_{21}\Psi_{21}\lambda_{62}$        $\lambda_{31}\Psi_{21}\lambda_{62}$        $\lambda_{42}\Psi_{22}\lambda_{62}$        $\lambda_{52}\Psi_{22}\lambda_{62}$        $\lambda_{62}\Psi_{22}\lambda_{62} + \theta_{66}$

- ☐ Indicator variances on the diagonal (in red)
- ☐ Cross-domain item correlations/covariances (in green) calculated via  $\Psi_{21}$

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## Covariance structure model

- ☐ Reformulate from the model specification
- ☐ This is the model used under the label **Covariance**

### Structure Analysis (COV-SEM)

$$\Sigma = \Lambda \Psi \Lambda^T + \Theta$$

- ☐ More generally, we assume that

$$\mathbf{Y} \sim N(\mathbf{0}, \Sigma)$$

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## Model identification

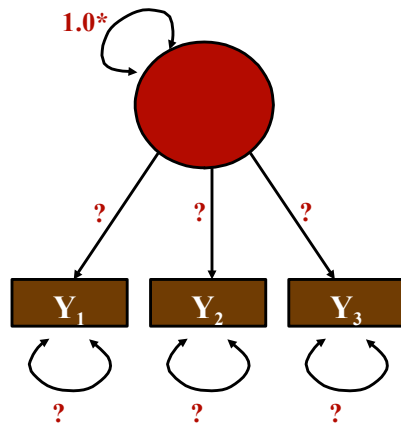
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## Model identification

- ☐ General idea: **everything is relative**
- ☐ Establish a point of reference
- ☐ Take measurement from this reference point
- ☐ Method of scale does not affect the model fit
- ☐ Instead, it affects parameter estimates and interpretation of constructs

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## Model identification



- ❑ Set a metric for the latent variable, we can say that the subjects scores come from  $N(0, 1)$ , i.e., fix latent variance ( $\psi_{11}$ ) = 1

## Model identification

- ❑ Use to set the scale of the latent construct
- ❑ Mean = 0, Variance = 1 for each common factor (recommended)
  - ❑ **Fixed Factor:** Fix latent variance (e.g.,  $\psi_{11} = 1$ )
  - ❑ **Marker Variable:** Fix one factor loading (e.g.,  $\lambda_{11} = 1$ )
- ❑ Mean = 0 for each unique factor (and error/residual)
- ❑ Common and unique factors are uncorrelated

## Bayesian analysis

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## Bayesian analysis

- ❑ Bayesian framework is more flexible than ML
- ❑ For instance, we can specify cross-loadings not fixed to 0, but with a priori like:

$$CI \sim N(0; 0.001)$$

- ❑ Challenging models to be estimated by ML are straightforward in Bayesian statistics, like interaction between factors

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## Bayesian CFA & SEM

- ☐ Bayesian estimation uses data augmentation
- ☐ We estimate the factor score ( $Z_{i1}$ ) for each subject, and use those scores as variables

$$Y_{i1} = \mu_1 + \lambda_{11}Z_{i1} + u_{i1}$$

- ☐ To estimate this model, we must impose identification conditions (like ML)
- ☐ Bayesian is more flexible; we do not need to assume that factors are normally distributed
- ☐ The posterior distribution of each parameter can be not normal

## Model fit and comparison



## Model fit - General

“With respect to model fit, researchers do not seem adequately sensitive to the fundamental reality that there is no true model..., that all models are wrong to some degree, even in the population, and that the best one can hope for is to identify a parsimonious, substantively meaningful model that fits observed data adequately well. At the same time, one must recognize that there may well be other models that fit the data to approximately the same degree... It is clear that a finding of good fit does not imply that a model is correct or true, but only plausible.”

MacCallum & Austin, 2000

## Model fit – There is no true model

- ☐ Models are parsimonious approximations of real-world phenomena
- ☐ Equivalent models may be just as appropriate
- ☐ Models with good fit are not necessarily meaningful
- ☐ “A finding of good fit does not imply that a model is correct or true, but only plausible” - MacCallum & Austin (2000)

## Ockam's Razor

- ☐ Models with fewer assumptions are to be preferred:
  - ☐ **Overfitting**: leads to poor predictions by learning too much from the data
  - ☐ **Underfitting**: leads to poor predictions by learning to little from data
- ☐ Many equivalent models can be specified
- ☐ You can have good model with no meaningful effects (model fit does not imply effect size)

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## Model fit - What prediction do we care about?

- ☐ What do we want our model to predict?
- ☐ Predicting observed data is easy and overestimates the model accuracy
- ☐ Out of sample prediction tests the accuracy of predicting observations that are not included in the model. True test of model performance
- ☐ LL is a measure of in-sample predictive accuracy

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## Model fit - Measures

- ☐ **PPP**

- ☐ Posterior Predictive P-value (PPP)

- ☐ Posterior Predictive Checks (PPC)

- ☐ **DIC** - Deviance Information Criterion

- ☐ **WAIC** - Watanabe-Akaike Information Criterion

- ☐ **LOO** - Leave-one-out (cross-validation, information criterion)

- ☐ **Bayes factors**

- ☐ **Approximate fit indices**

- ☐ Posterior distribution of fit indices

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## PPP – Posterior predictive p-values

- ☐ Measure of the model's absolute fit

- ☐ Compares observed likelihood ratio test (LRT) statistics to likelihood ratio test statistics generated from the model's posterior predictive distribution:

1. Compute the observed LRT statistic
2. Generate artificial data from the model
3. Compute the posterior predictive LRT of the artificial data
4. Record which LRT is higher

- ☐ The PPP is the proportion of times the posterior LRT is larger

- ☐ Perfect fit is 0.5

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## DIC

- ☐ Deviance Information Criterion
- ☒ **Goal: measures the model's predictive accuracy**
- ☐ Model comparison

$$\text{DIC} = -2\text{LL} + 2\text{efp}$$

- ☐ It penalizes the LL for the effective number of parameters
- ☐ Lower values indicate better fit

## WAIC

- ☐ Watanabe-Akaike Information Criterion
- ☒ **Goal: measures the model's predictive accuracy**
- ☐ Model comparison

$$\text{WAIC} = -2\text{lppd} + 2\text{efp}$$

- ☐ Uses the casewise LL (lppd), instead of the overall LL model like DIC
- ☐ As DIC, it penalizes the LL by the effective number of parameters
- ☐ Lower values indicate better fit

## LOO

- ☐ Leave-one-out (cross-validation, information criterion)
- ☐ Asymptotically equivalent to WAIC
- ☐ Predictive density of each individual observation from a cross-validation standpoint; that is, the predictive density when we hold out one observation at a time and use the remaining observations to update the prior
- ☐ Lower values indicate better fit

## LOO and WAIC

- ☐ Unbiased estimates of the predictive ability of a given model
- ☐ Approximate cross-validation predictive ability
- ☐ The R package loo allows to compare models based on LOO and WAIC

## Bayes factors

- ❑ Between two models is defined as the ratio of the models marginal likelihood

$$BF = f(Y | M_1) / f(Y | M_2)$$

- ❑ Values greater than 1 favor model 1, values lower than 1 favor model 2
- ❑ Blavaan compares models with LOO, WAIC, and log-Bayes Factor using the function blavCompare

## Model fit in CFA & SEM

## Posterior predictive checks

- ❑ As general process can be done with any meaningful statistic (see lecture on PPP)
- ❑ In SEM, we use the LRT as a measure of multivariate deviation
- ❑ Generally, it is a likelihood ratio test (LRT): comparing the log-likelihood for the hypothesis model (implied) to the saturated model (observed)

$$\chi_{ML}^2 = -2(\ell_H - \ell_S)$$

## Posterior predictive checks

- ❑ Bayesian versions of relative and absolute fit measures, commonly used in frequentist SEM
- ❑ Like: RMSEA, gamma-hat, CFI, TLI
- ❑ Allowing to interpret the fit of the model relative to the best and worst fitting models
- ❑ Generate posterior distributions for each fit index quantifying the uncertainty
- ❑ The mean of these posteriors are equivalent to the frequentist fit index (under diffuse priors)

## SEM model fit

### ❑ Root Mean-Squared Error of Approximation (RMSEA)

❑ Good fit: RMSEA < 0.05

### ❑ Standardized Root Mean Square Residual (SRMR)

❑ Good fit: SRMR < 0.05

### ❑ Gamma-hat ( $\hat{\Gamma}$ )

❑ Acceptable fit: Gamma-hat > 0.9

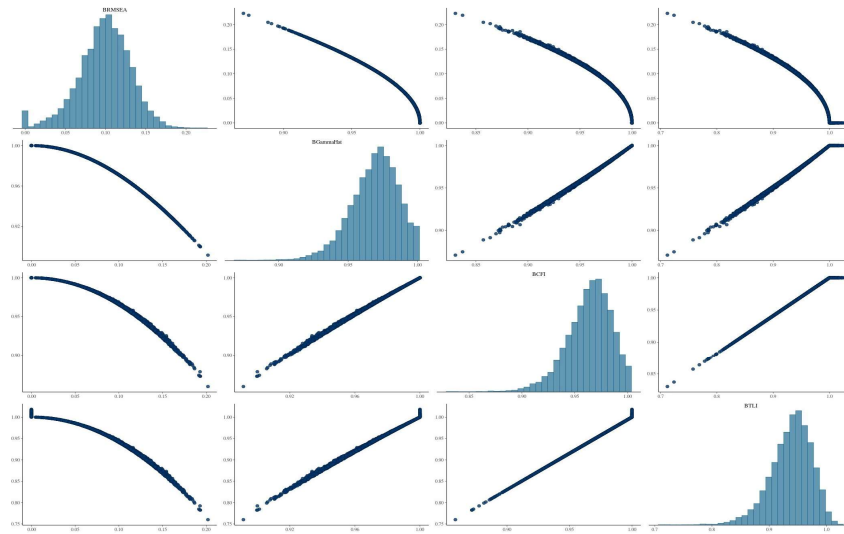
### ❑ Tucker Lewis' Index (TLI)

❑ Acceptable fit: TLI > 0.9

### ❑ Comparative Fit Index (CFI)

❑ Acceptable fit: CFI > 0.9

## Summary overall fit measures





## Summary overall fit measures

- ❑ Evaluate the “overall” discrepancy between the model and the observed covariance
- ❑ Report: gamma-hat, CFI, and SRMR, which are less sensitive to characteristic other than misfit
- ❑ Others can be RMSEA (sensitive to model size) or TLI (sensitive to model type)
- ❑ Common interpretation should be treated as general guidelines, not strict cutoffs

## Local fit

## Local fit

- ❑ Overall measures can miss misfit for a specific indicator
- ❑ Can find “acceptable” overall fit and have local misfit
- ❑ Local fit helps find if we have missed something for a single indicator:
  - ❑ Residual correlations
  - ❑ Modification indices

## Residual correlations

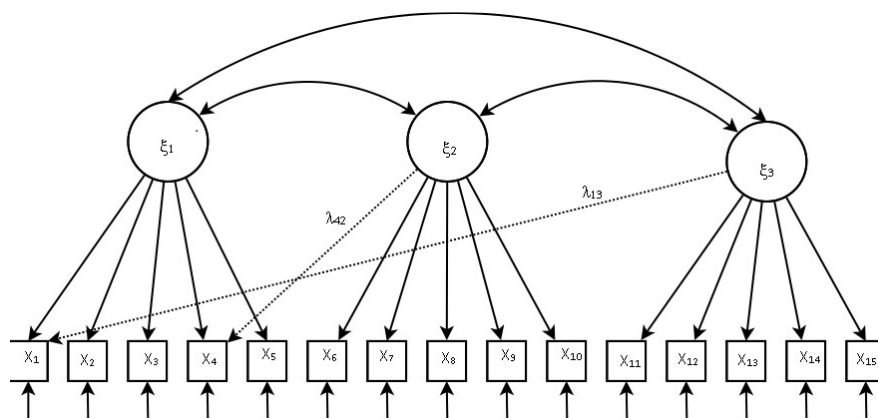
- ❑ Helps identify for which variables might be misfit, but doesn't say how to fix it
- ❑ Might need to add parameters to correct this
- ❑ Common interpretation:  $|r| > 0.10$  is possible evidence of poor local fit
- ❑ Posterior distribution of residual correlations (evaluate the degree and uncertainty of the deviation)
- ❑ Model modifications should be theoretically defended

## Modification indices

- ❑ Indicates the approximate improvement in model fit ( $\chi^2$ ) if a parameter is added
- ❑ When added one parameter at a time and evaluate model fit (overall and local) at each step
- ❑ ONLY add parameters that you can defend theoretically
- ❑ Posterior distribution of modification indices
- ❑ Use mi to identify the parameters that would have the largest impact on model fit

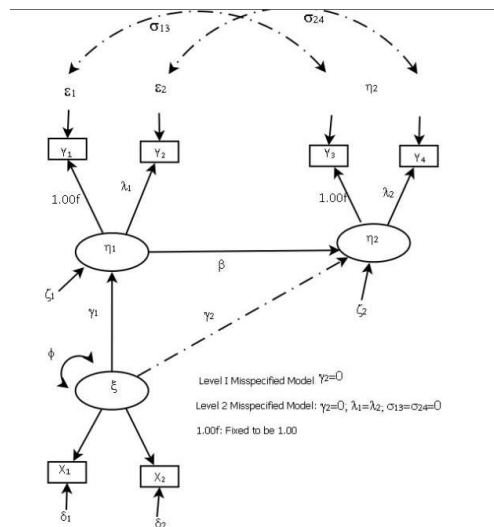
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## Local fit



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## Residual correlations



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## Be careful!!

- ☐ With modification indices and residual correlations it's easy to add parameters that would improve the model fit
- ☐ If you keep adding parameters to your model it will achieve "good" fit
- ☐ This can lead to models with "good" fit but no meaningful interpretation
- ☐ Remember theory!!!

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## Modification indices

In case you want to modify the model in lavaan:

- ❑ **Cross-loading:**  $F2 \sim V1$  = part of the reliable information in V1 is related to F1 (original factor) and another part is related to F2
- ❑ **Residual correlation:**  $V1 \sim V2$  = there is something in common between V1 and V2, above and beyond what is explained by the underlying factors